

Large Language Models Do Multi-Label Classification Differently

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What is Multi-Label Classification?

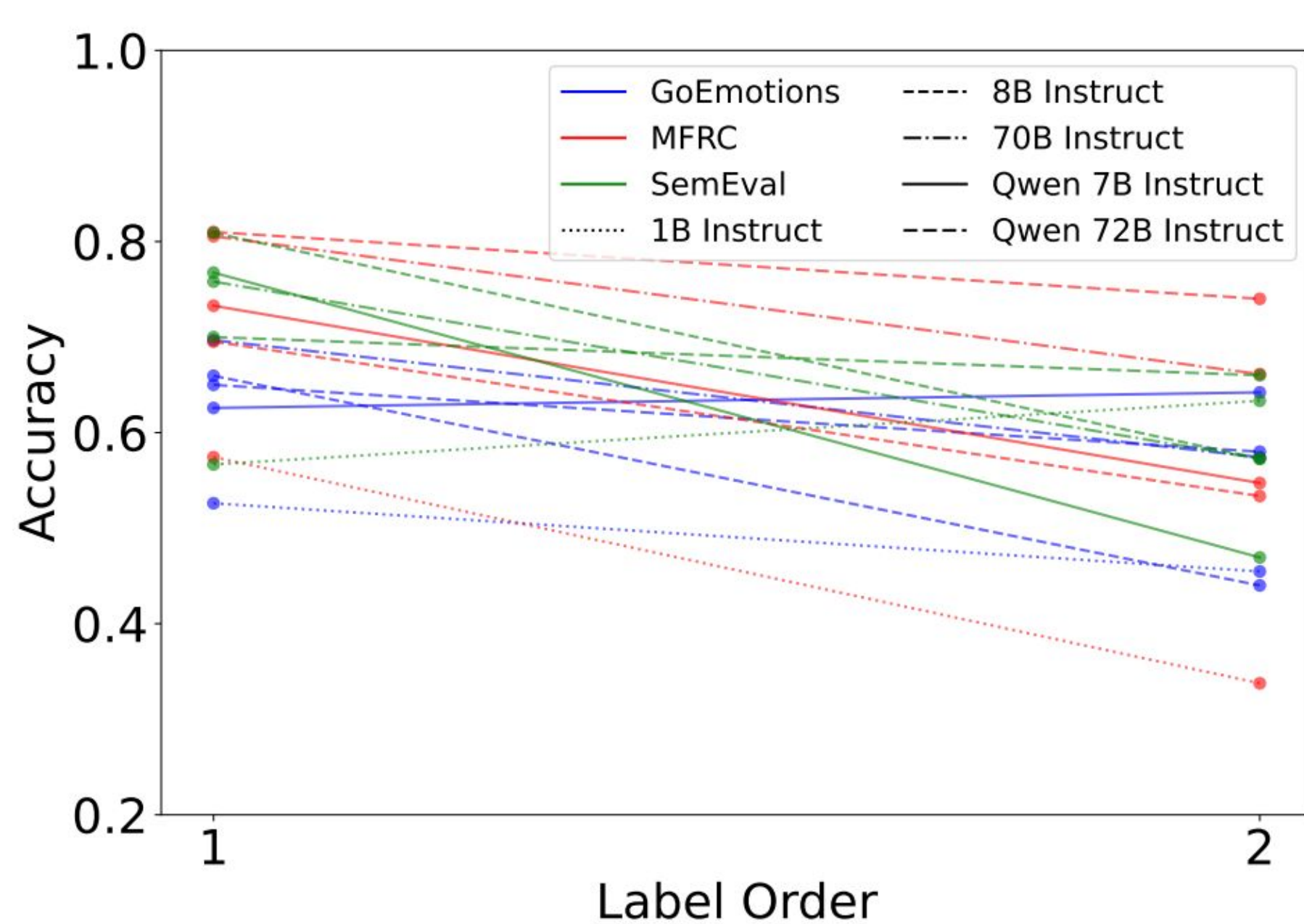
Multi-label classification, where every label's probability is independent from every other label's, differs from single-label settings since summing all probabilities will not equal one. LLMs are pretrained under next-token prediction, a single-label classification task, which we hypothesize partially explains some of the interesting phenomena we observe.

Datasets

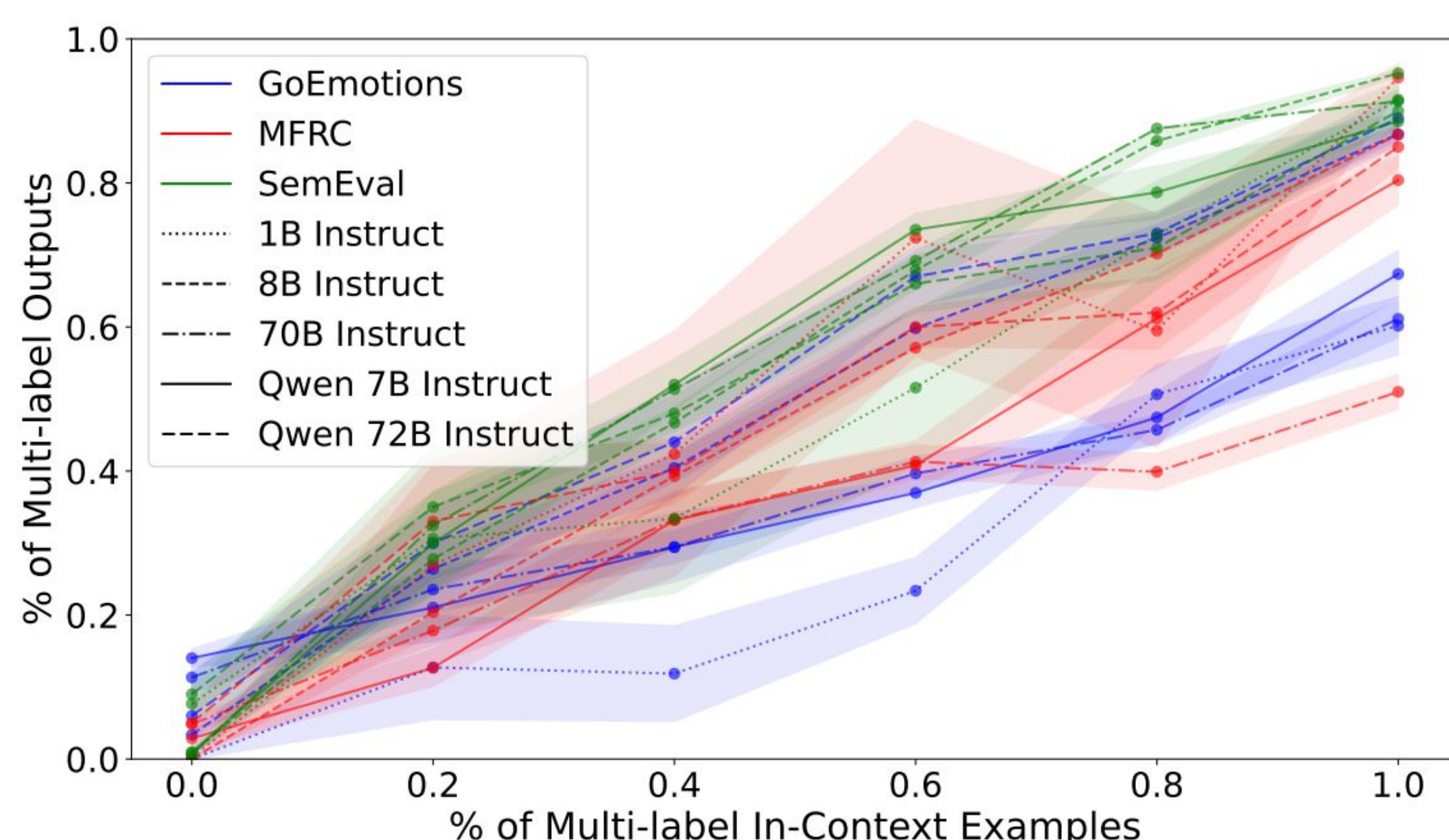
Dataset	Annotators (per example)	Cohen's Kappa	0 labels	1 label	2 labels	3+ labels
GoEmotions	81 (3.58)	0.27	29%	62%	8%	1%
MFRC	6 (2.99)	0.21	78%	18%	3%	<1%
SemEval	— (—)	—	1%	13%	40%	46%

Table 1: Annotation statistics and label distributions. The public release of SemEval does not include individual annotator labels, only aggregates.

Labels Are More Accurate the Earlier They Are Predicted



In-Context Examples Help LLMs Predict Multiple Labels More

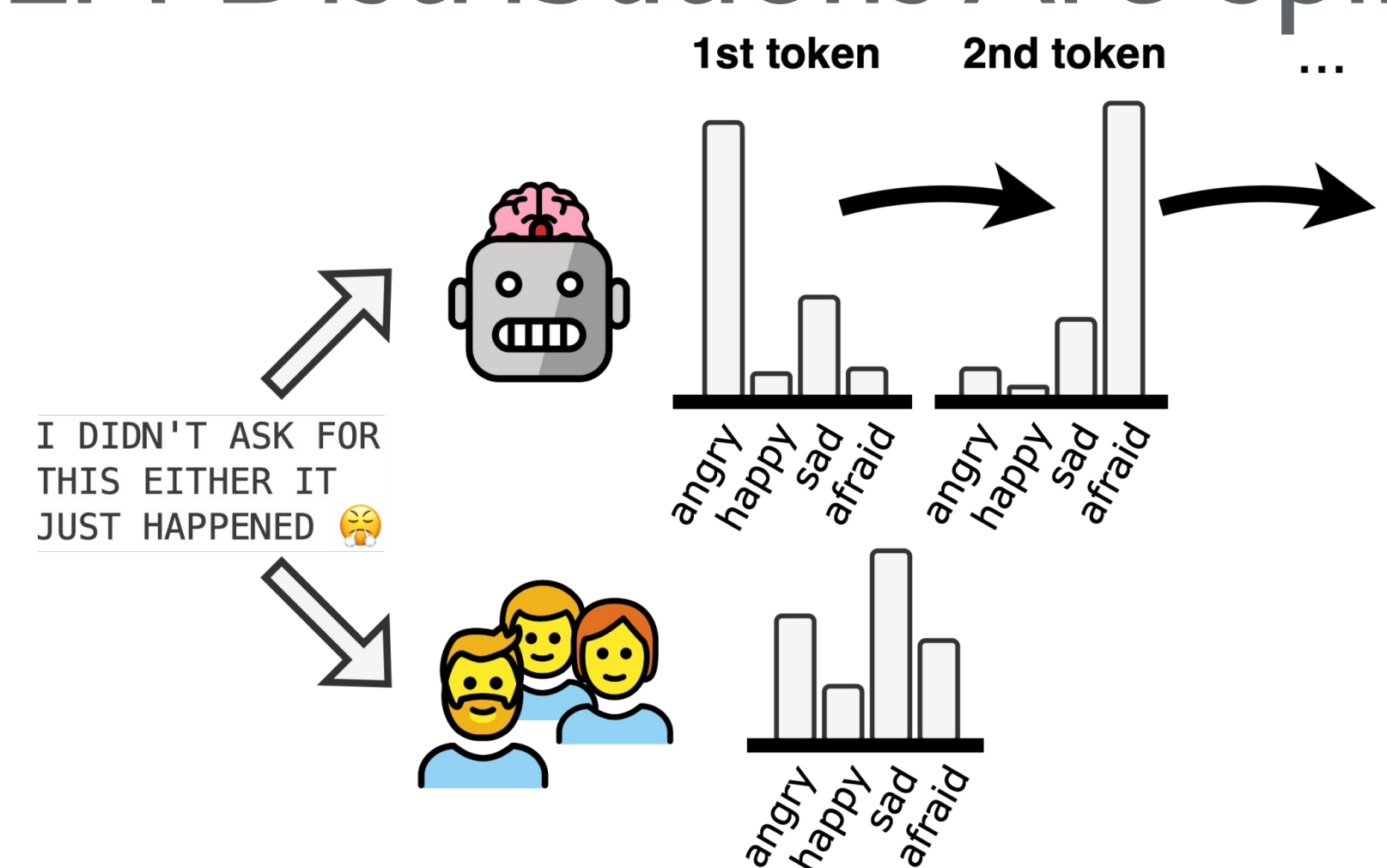


We found the chance of predicting multiple labels was close to the frequency of multi-label prompt examples.

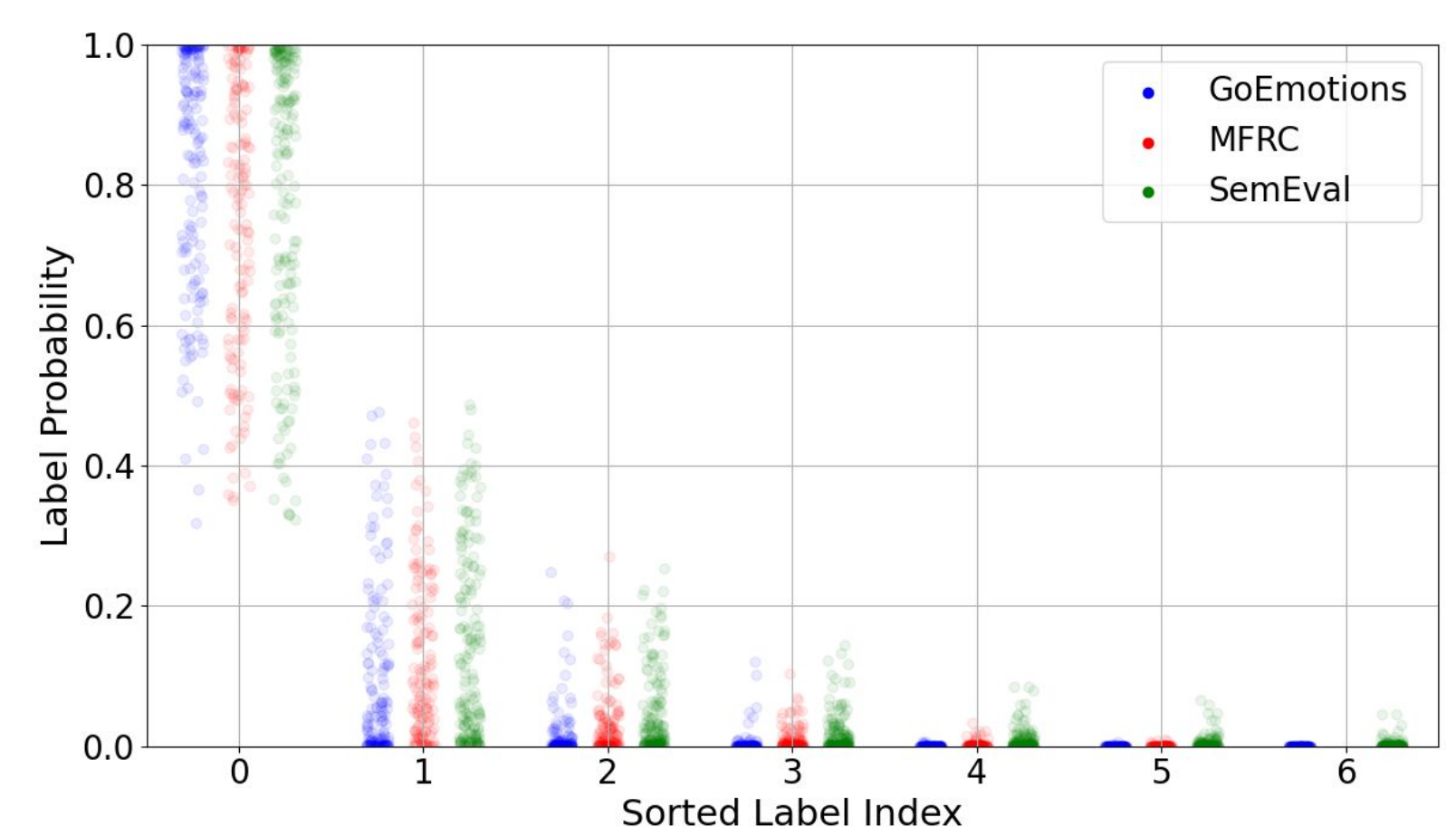
Abstract

Large Language Models (LLMs) are not well understood in multi-label classification, and we dedicate this work to studying their behavior. We find several interesting findings: when generating multiple labels, first labels' probability distributions often diverge from the final output; models suppress all but one label at each step; models lack stopping criterion when predicting intermediate labels; and relative ranking of non-predicted labels is not consistent with actual predictions.

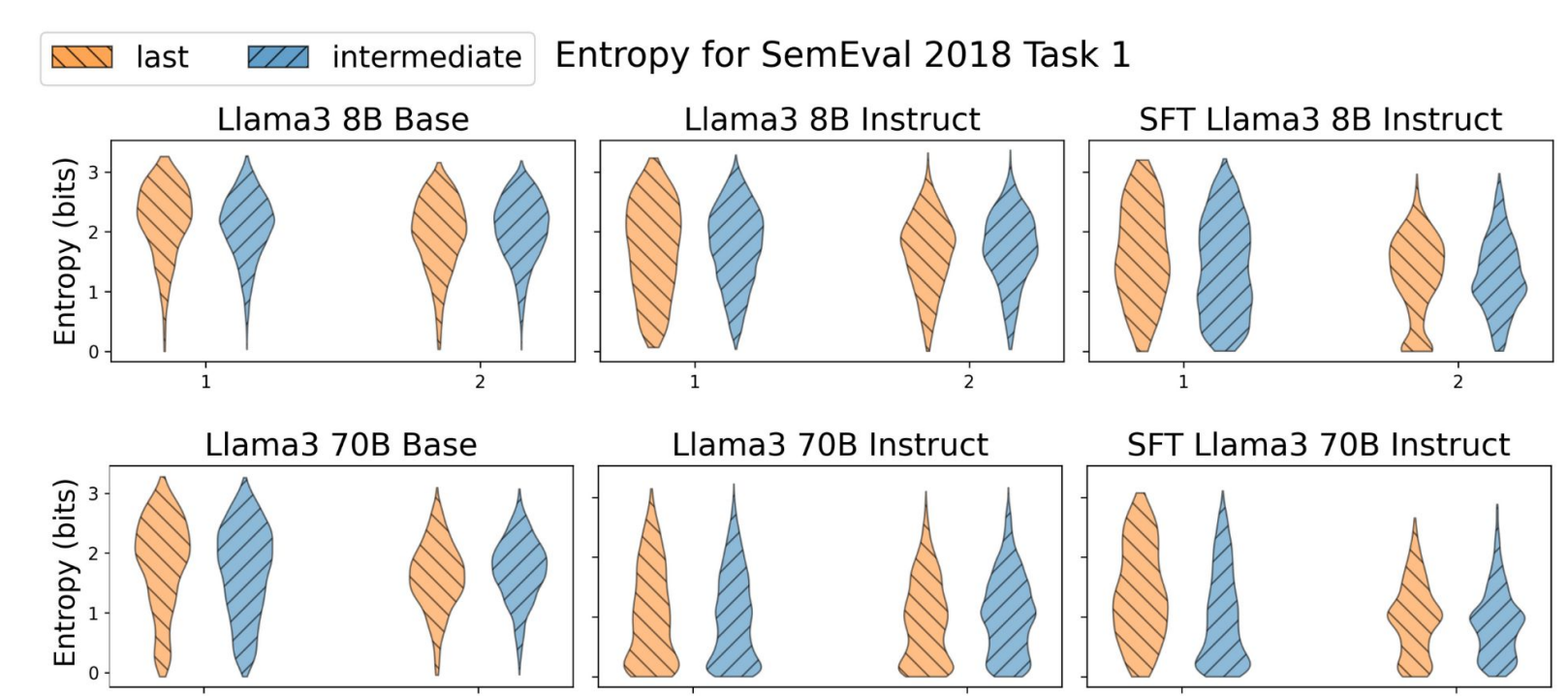
LLM Distributions Are Spiky



LLMs tend to predict labels one-by-one with very high confidence instead of a holistic probability distribution over all labels, even for labels it will predict later on.



LLMs predict sequentially spiky probability distributions, where multiple-label outputs are generated by sequentially predicting individual labels with high probability. We find the relative ranking between yet-to-be-predicted labels often does not line up with what is actually predicted.



Our observations are supported by the entropy at each generation step (intermediate versus last):

1. **Sequential Spikiness**
2. **Lack of Stopping Criterion:** last predictions and intermediate predictions show the same behavior
3. **Spurious Relative Ranking:** The second highest label is only ~50% likely to be predicted next